Direct Displacement-Based Design (DDBD)

of an N-Storey MDOF Shear Building with Nonlinear Time-History Analysis in OpenSeesPy

Complete Code Explanation and All Formulas (MDOF Example)

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1. Overview

This document explains the MDOF extension of the DDBD script (MDOF_DIRECT_DISPLACEMENT-BASED_DESIGN.py). A 10-storey shear building is designed with the Direct Displacement-Based Design procedure of Priestley, Calvi and Kowalsky (2007) and then verified with a nonlinear dynamic time-history analysis (THA) in OpenSeesPy. Unlike the SDOF version, the design now starts from a displacement profile of the whole building, which is reduced to an equivalent SDOF (substitute) structure, and the resulting base shear is distributed back to the storeys.

The workflow has seven stages:

- Design displacement profile Δi (inelastic shape and higher-mode drift reduction ωθ).
- Substitute (equivalent SDOF) structure: Δd, me, He.
- Yield displacement Δy = θy·He, ductility μ and equivalent damping ξeq.
- Effective period Teff from the ξeq-damped design displacement spectrum at Δd.
- Effective stiffness Ke and design base shear Vb.
- Distribution of Vb to the floors, storey shears Vi and storey strength factors Ci of a shared backbone.
- Verification by nonlinear THA with a record scaled to the design spectrum at Teff.

The case study uses a bilinear (linear up to the corner period, then constant) design displacement spectrum, which can be replaced by any code spectrum (e.g., Eurocode 8 or NZS 1170.5).

1.1 Files and Their Roles

File	Role
MDOF_DIRECT_DISPLACEMENT-BASED_DESIGN.py	Main script: DDBD design steps 1–6, ground-motion scaling, call to the THA, verification and plots.
MDOF_DYNAMIC_TIME_HISTORY.py	Shear-building library: peak-oriented hysteresis, modal properties, Rayleigh coefficients, Newmark–Newton numerical solver and the OpenSeesPy model (MODAL_PROPERTIES, MDOF_OPENSEES_DYNAMIC).
ANALYSIS_FUNCTION.py	Fallback routine that cycles through test/algorithm combinations when a transient step does not converge.
PLOT_FORCE_DISP_FUN.py	Prints and plots the elastic, plastic and collapse (secant) stiffnesses of the backbone.
Ground_Acceleration_X.txt	Acceleration record in g, dt = 0.01 s.

2. Building and Design Parameters

All quantities use N, m, kg and s. The building is a regular shear building: one lateral degree of freedom per floor and one nonlinear spring per storey.

Symbol	Value	Meaning
N	10	Number of storeys
hs	3.0 m	Constant storey height; floor heights Hi = hs·i, roof Hn = 30 m
mi	200 × 10³ kg	Floor mass (same at every floor; can be edited)
θc	0.025	Code / performance drift limit at the critical (first) storey
θy	0.010	Storey yield drift (RC frame: 0.5·εy·Lb/hb)
Ft/Vb	0.10	Fraction of Vb applied as an extra force at the roof
ξel	0.05	Elastic viscous damping (spectrum base and THA Rayleigh)
Sd,max	0.90 m	Plateau of the 5 % displacement spectrum
TD	5.0 s	Corner period of the spectrum

3. Design Displacement Profile

3.1 Inelastic Shape

The normalised inelastic mode shape depends on the number of storeys:

$$\delta i = Hi / Hn \qquad for\ N \le 4$$

$$\delta i = (4/3)\cdot(Hi/Hn)\cdot(1 - Hi/(4\cdot Hn)) \qquad for\ N > 4$$

3.2 Higher-Mode Drift Reduction

Higher modes amplify the drift of tall buildings, so the design drift of the critical storey is reduced (Hn in metres):

$$\omega\theta = min(\,1.0\,,\ 1.15 - 0.0034\cdot Hn\,)$$

$$\theta d = \omega\theta \cdot \theta c$$

For Hn = 30 m, 1.15 − 0.0034·30 = 1.048 > 1, so ωθ = 1.0 and θd = 2.5 %.

3.3 Design Displacements and Drifts

The shape is scaled so that the first storey (critical storey) reaches the design drift:

$$\Delta i = \delta i \cdot (\theta d \cdot H1) / \delta 1$$

$$drift_i = \Delta i - \Delta i{-}1 \quad (\Delta 0 = 0)$$

Because the first storey is critical, its drift is exactly θd = 2.5 %; the upper-storey drifts decrease with height (1.35 % at the roof storey in this example).

4. Substitute (Equivalent SDOF) Structure

4.1 Design Displacement, Effective Mass and Height

$$\Delta d = \Sigma\, mi \cdot \Delta i^2 \,/\, \Sigma\, mi \cdot \Delta i$$

$$me = \Sigma\, mi \cdot \Delta i \,/\, \Delta d$$

$$He = \Sigma\, mi \cdot \Delta i \cdot Hi \,/\, \Sigma\, mi \cdot \Delta i$$

4.2 Yield Displacement and Ductility

In the SDOF script the yield displacement was an input. Here it follows from the yield drift and the effective height:

$$\Delta y = \theta y \cdot He$$

$$\mu = \Delta d \,/\, \Delta y$$

4.3 Equivalent Viscous Damping

For RC frames the Takeda-thin rule is used. Note the factor π in the denominator, which is part of the Priestley expression and is present in the script:

$$\xi eq = \xi el + 0.565 \cdot (\mu - 1) \,/\, (\mu \cdot \pi) \qquad for\ \mu > 1,\ otherwise\ \xi eq = \xi el$$

The first term is the inherent elastic damping; the second term is the hysteretic contribution that grows with ductility.

5. Design Displacement Spectrum and Effective Period

5.1 Damping Modification Factor

$$\eta = 1.0 \qquad if\ \xi \le 0.05$$

$$\eta = max(\sqrt{[10 \,/\, (5 + 100\xi)]}, 0.55) \qquad if\ \xi > 0.05$$

5.2 5 % Damped Spectrum

$$Sd,5\%(T) = Sd,max \cdot min(T, TD) \,/\, TD$$

The spectrum rises linearly to the corner period TD = 5 s and is constant afterwards (Sd,max = 0.90 m).

5.3 Damped Spectrum

$$Sd(T, \xi) = \eta(\xi) \cdot Sd,5\%(T)$$

5.4 Effective Period

Teff is the period at which the ξeq-damped spectrum equals Δd. Because the target lies on the linear branch, the equation has a closed-form solution and no numerical solver (fsolve) is needed:

$$Teff = \Delta d \cdot TD \,/\, (\eta(\xi eq) \cdot Sd,max)$$

If Δd > η·Sd,max the target lies above the plateau and no Teff exists; the script raises a ValueError and asks the user to reduce the drift limit or increase Sd,max.

6. Design Base Shear and Force Distribution

6.1 Effective Stiffness and Base Shear

$$Ke = 4\pi^2 \cdot me / Teff^2$$

$$Vb = Ke \cdot \Delta d$$

P-Δ effects are ignored (Priestley: Vb − C·P·Δd/He).

6.2 Floor Forces

A fraction Ft = 0.10·Vb is placed at the roof and the remainder is distributed in proportion to mi·Δi:

$$Fi = (1 - Ft/Vb) \cdot Vb \cdot mi{\cdot}\Delta i / \Sigma\, mj{\cdot}\Delta j \quad (+ Ft \text{ at the roof, } i = N)$$

6.3 Storey Shears

$$Vi = \Sigma(j \geq i)\, Fj$$

In the code this is the reversed cumulative sum of the floor forces, so V1 equals Vb.

7. Storey Backbones and Strength Scaling

7.1 Shared Multilinear Backbone

All storeys share one four-point backbone expressed in storey drift. With a reference yield force Fy,ref = 1000 N and a storey yield drift Dy = θy·hs = 0.03 m:

$$Force = Fy,ref \cdot [1.0,\ 1.2,\ 0.25,\ 0.10]$$

$$Displacement = Dy \cdot [1,\ 10,\ 12,\ 15]$$

These are the yield, peak (1.2·Fy at 10·Dy), first residual (Fy/4 at 12·Dy) and second residual (Fy/10 at 15·Dy) points. PLOT_FORCE_DISP_FUN reports the elastic stiffness Ke = F1/D1, the plastic (secant to the second point) stiffness and the collapse (secant to the last point) stiffness, and the tangent stiffness at every point.

7.2 Storey Strength Factors Ci

Each storey spring is the shared backbone multiplied by a factor Ci (stiffness and strength scale together). Two options are available through the flag DESIGN_ON_BACKBONE:

- **True (default):** the storey shear Vi must be reached at the design drift of storey i, which is consistent with the secant-stiffness basis of DDBD:

$$Ci = Vi / F_backbone(drift_i)$$

where F_backbone is obtained by linear interpolation (np.interp) on the backbone points, including the origin.

- **False:** Vi is used directly as the yield strength (same convention as the SDOF script):

$$Ci = Vi / Fy,ref$$

Initial storey stiffness is then ki = Ci · F1/D1, which makes the lower storeys stiffer than the upper ones.

8. Ground-Motion Scaling

The record is scaled so that its linear response spectrum, at the effective period and the equivalent damping, equals the design displacement at Teff, i.e. the demand assumed in the design:

$$SF = \Delta d \,/\, Sd,record(Teff, \xi eq)$$

$$GM_FACTOR = g \cdot SF$$

Sd,record is computed by SD_LINEAR, a linear SDOF solved with the Newmark average-acceleration method (γ = 0.5, β = 0.25); its peak absolute displacement is the spectral displacement. The raw record is in g, so g = 9.81 m/s² converts it to m/s². With AUTO_SCALE = False the user-defined GM_SCALE is used instead.

9. OpenSees Model and Nonlinear Time-History Analysis

9.1 Equation of Motion

$$M \cdot \ddot{u} + C \cdot \dot{u} + f_int(u, history) = -M \cdot r \cdot ag(t) \qquad r = \{1, \ldots, 1\}$$

u are floor displacements relative to the ground. Storey i acts on the drift di = ui − ui−1 (u0 = 0) and the internal force at floor i is Vi − Vi+1, giving the tridiagonal tangent stiffness assembled by ASSEMBLE_K.

9.2 Geometry, Mass and Story Springs

- A 1-D model (ndm = 1, ndf = 1) with nodes 0 to N; node 0 is fixed and nodes 1 to N carry the floor masses.
- Each storey i is a zeroLength element between nodes i−1 and i, with a HystereticSM material whose force ordinates are scaled by Ci while the displacement ordinates are unchanged:

  $$pos = [Ci \cdot F1, D1, Ci \cdot F2, D2, Ci \cdot F3, D3, Ci \cdot F4, D4], \quad neg = [-Ci \cdot F1, -D1, \ldots]$$

- The pinching and damage parameters are set to 1.0, 1.0, 0, 0, 0 (no pinching, no damage), which corresponds to the peak-oriented hysteresis rule used in the Python solver (unloading with the initial stiffness k0).
- The option -doRayleigh 1 is required, because zeroLength elements ignore the stiffness-proportional Rayleigh term by default.

9.3 Modal Properties

The initial elastic structure (K0, M) is solved as a symmetric eigenproblem:

$$(M\text{^}(-1/2) \cdot K0 \cdot M\text{^}(-1/2)) \cdot \tilde{\varphi} = \omega^2 \cdot \tilde{\varphi}, \quad \varphi = M\text{^}(-1/2) \cdot \tilde{\varphi}$$

The script prints the first three periods of the designed building, which indicates how far the elastic period is from Teff.

9.4 Rayleigh Damping

In the MDOF model the damping is no longer mass-proportional only; it is calibrated to ξ in two modes (default modes 1 and 2):

$$C = a0 \cdot M + a1 \cdot K0$$

$$a0 = 2\xi \cdot \omega i \cdot \omega j \,/\, (\omega i + \omega j), \quad a1 = 2\xi \,/\, (\omega i + \omega j)$$

OpenSees: ops.rayleigh(a0, 0, a1, 0), where the third argument (betaKinit) applies the initial stiffness, giving a constant C identical to the Python solver. IMPORTANT: the THA uses the elastic

ξel = 5 % only. The hysteretic energy dissipation is already contained in the HystereticSM springs, so using ξeq here would count the damping twice. (In the SDOF script ξeq was used for the mass-proportional term.)

9.5 Ground Motion and Analysis

A Path time series is created from the record with -factor GM_FACTOR, and a UniformExcitation pattern applies it in direction 1. The analysis uses BandGeneral, Plain numberer and constraints, a NormDispIncr test (tolerance 10^{-6}, 500 iterations), the Newton algorithm, the Newmark integrator (γ = 0.5, β = 0.25) and the Transient analysis. Each step calls ops.analyze(1, dt); if it fails, ANALYSIS() cycles through test/algorithm combinations (NormDispIncr, RelativeEnergyIncr, RelativeNormUnbalance, …; KrylovNewton, SecantNewton, RaphsonNewton, PeriodicNewton, BFGS, Broyden, NewtonLineSearch) until a step converges.

At every step the script stores floor displacements, the base reaction (ops.nodeReaction) and the storey shears (the negative element force of each zeroLength element).

10. Post-Processing and Verification

10.1 Envelopes

$$U_env,i = max\,|ui(t)|\,,\quad D_env,i = max\,|di(t)|\,,\quad V_env,i = max\,|Vi(t)|$$

$$\Delta d,THA = \Sigma\, mi \cdot U_env,i^{2} \,/\, \Sigma\, mi \cdot U_env,i$$

10.2 Printed Checks

- Roof displacement: THA envelope against the design roof displacement Δn (ratio).
- Equivalent design displacement: Δd,THA against Δd (ratio).
- Maximum storey drift: THA against the limit θc, with the governing storey.
- Base shear: THA V1 against the design Vb (ratio).

Ratios near 1.0 indicate that the DDBD strength, the damping model and the record scaling are consistent. Values well above 1 for drift suggest a soft-storey concentration or a record that is stronger than the design spectrum at periods away from Teff.

10.3 Plots

Eight panels are produced: (1) scaled ground acceleration; (2) roof displacement with the ±Δn design lines; (3) global hysteresis (roof displacement vs. base reaction); (4) design displacement spectra with the (Teff, Δd) point; (5) displacement profile, design vs. THA envelope; (6) storey drift profile with the drift limit; (7) storey shear profile, design Vi vs. THA envelope; (8) base shear time history with ±Vb.

11. Worked Example (Numbers from the Script)

Evaluating the design steps for the parameters of Section 2 gives the following values (the THA results depend on the OpenSeesPy run and are not listed).

Quantity	Value	Comment
ωθ / θd	1.000 / 2.50 %	No higher-mode reduction at Hn = 30 m
Δd	0.423 m	Equivalent SDOF displacement

Quantity	Value	Comment
me	1 649 864 kg	82.5 % of total mass (2 000 000 kg)
He	20.45 m	68 % of Hn
Δy	0.2045 m	θy · He
μ	2.07	Δd / Δy
ξeq	14.3 %	η = 0.720
Teff	3.265 s	Δd · TD / (η · Sd,max)
Ke	6.11 × 10⁶ N/m	4π²·me / Teff²
Vb	2.586 × 10⁶ N	13.2 % of the building weight

Floor and storey results (N, m):

Storey	Δi [m]	Drift [%]	Fi [N]	Vi [N]	Ci
1	0.0750	2.500	50,002	2,585,592	2502.2
2	0.1462	2.372	97,440	2,535,590	2460.6
3	0.2135	2.244	142,314	2,438,149	2372.6
4	0.2769	2.115	184,624	2,295,835	2240.3
5	0.3365	1.987	224,370	2,111,211	2065.9
6	0.3923	1.859	261,551	1,886,841	1851.5
7	0.4442	1.731	296,168	1,625,290	1599.3
8	0.4923	1.603	328,221	1,329,123	1311.6
9	0.5365	1.474	357,709	1,000,902	990.5
10	0.5769	1.346	643,193	643,193	638.3

The roof force includes the extra Ft = 0.10·Vb = 258 559 N. Because the backbone is entered at the design drift of each storey (DESIGN_ON_BACKBONE = True), Ci is slightly smaller than Vi/Fy,ref; for example C1 = 2 585 592 / 1 033.3 ≈ 2502 where 1033.3 N is the backbone force at the 2.5 % drift.

12. Simplifications and Limitations

- P-Δ is ignored (no Vb reduction and no geometric nonlinearity in the model).
- Constant storey height and the same yield drift θy in every storey.
- Shear-building idealisation: one DOF per floor, no axial or beam-flexibility effects.
- Rayleigh damping in the THA is the elastic 5 % only; ξeq is used only to derive Teff and to scale the record.
- A single record scaled at one period; for design verification a suite of records (e.g., 7) should be used and the mean response compared with the targets.
- Design spectrum is bilinear and illustrative; replace SD_5PCT() and DAMPING_FACTOR() with the code spectrum when required.

13. Summary of All Formulas

Quantity	Formula
Inelastic shape (N > 4)	$\delta i = (4/3)(Hi/Hn)(1 - Hi/(4Hn))$
Drift reduction	$\omega\theta = min(1, 1.15 - 0.0034 \cdot Hn);\ \theta d = \omega\theta \cdot \theta c$
Design profile	$\Delta i = \delta i \cdot (\theta d \cdot H1)/\delta 1$
Design displacement	$\Delta d = \Sigma\, mi \cdot \Delta i^2 / \Sigma\, mi \cdot \Delta i$
Effective mass	$me = \Sigma\, mi \cdot \Delta i / \Delta d$
Effective height	$He = \Sigma\, mi \cdot \Delta i \cdot Hi / \Sigma\, mi \cdot \Delta i$
Yield displacement	$\Delta y = \theta y \cdot He$
Ductility	$\mu = \Delta d / \Delta y$
Equivalent damping	$\xi eq = \xi el + 0.565(\mu - 1)/(\mu\pi)$
Damping factor η	$\eta = max(\sqrt{[10/(5+100\xi)]}, 0.55)\ \ (\xi > 0.05)$
5 % spectrum	$Sd,5\%(T) = Sd,max \cdot min(T, TD)/TD$
Damped spectrum	$Sd(T, \xi) = \eta(\xi) \cdot Sd,5\%(T)$
Effective period	$Teff = \Delta d \cdot TD / (\eta \cdot Sd,max)$
Effective stiffness	$Ke = 4\pi^2 \cdot me / Teff^2$
Design base shear	$Vb = Ke \cdot \Delta d$
Floor forces	$Fi = (1 - Ft/Vb) \cdot Vb \cdot mi \cdot \Delta i / \Sigma\, mj \cdot \Delta j\ (+Ft\ at\ roof)$
Storey shears	$Vi = \Sigma(j \geq i)\ Fj$
Strength factors	$Ci = Vi / F_backbone(drift_i)\ \ or\ \ Ci = Vi / Fy,ref$
Record scale factor	$SF = \Delta d / Sd,record(Teff, \xi eq)$
Rayleigh coefficients	$a0 = 2\xi \cdot \omega i \cdot \omega j/(\omega i + \omega j);\ a1 = 2\xi/(\omega i + \omega j);\ C = a0 \cdot M + a1 \cdot K0$
Equivalent THA Δd	$\Delta d,THA = \Sigma\, mi \cdot U_env,i^2 / \Sigma\, mi \cdot U_env,i$

14. What Changes from the SDOF Version

Item	SDOF script	MDOF script
Target	Δd prescribed directly	Δi profile from θc, ωθ; Δd, me, He derived
Yield displacement	Input (Δy,est)	θy · He
Mass	From T0 and K0	Given floor masses; me derived
Effective period	fsolve on the spectrum	Closed form on the linear branch
Strength	Backbone scaled by Vb/Fy	Vb distributed to storeys; one Ci per storey
Damping in THA	Mass-proportional with ξeq	Rayleigh (2 modes) with elastic ξel = 5 %
Elements	One node, parallel springs	N storey zeroLength springs in series
Record scaling	Fixed GM_SCALE	Auto-scaled to Sd(Teff, ξeq) = Δd
Verification	Peak displacement vs. Δd	Profiles of displacement, drift, storey shear and Vb

15. References

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30 Years of Direct Displacement-Based Design: Where We Started, What's Been Done, Where We're Going (WCEE 2024).

OpenSees / OpenSeesPy documentation: HystereticSM material, zeroLength element, rayleigh, Newmark integrator, UniformExcitation, Path time series.